

weed-control, autonomous mowing, spraying and thinning, sorting and utility. Autonomous robots are being developed to harvest crops at a higher volume and faster pace than human labourers. For example, Harvest Croo has developed an autonomous strawberry picking machine, while Abundant Robotics has developed a robot to pick apples from trees.

Researchers at University of Cambridge have developed a vegetable picking robot called Vegebot, as shown in Fig. 3. It is an autonomous iceberg-lettuce-harvesting robot that comprises a laptop computer, a standard six-degree-of-freedom (DOF) UR10 robot arm, two cameras and a custom end effector, all housed on a mobile platform.

Robotics dairy farming. Modern dairy farming techniques that employ robotics is quite interesting. Till now, milking cows has been done by humans using hands. Now in the 21st century, cows have been taught to adapt to new environments and techniques including robotic systems. They can be trained to milk two to three times a day using the new systems. Some automatic robotic operations allow milking more than 600 cows at a time.

Robotic milking systems and rotary parlours have been around for a few years, but a combination of the



Fig. 3: Vegebot, an autonomous iceberg-lettuce-harvesting robot (Credit: <https://hackaday.com>)

two concepts into one system is new. The world's first robotic rotary dairy developed by DeLaval was unveiled in 2010, and the system has become popular in many countries since then. Each robotic module in the automatic milking system handles every step of the milking process including teat cup attachment, pre-dipping, pre-milking, milking and post-dipping in one single attachment. The prototype of the robotic rotary automatic milking system developed by DeLaval is shown in Fig. 4.

AI-based farm management. An AI-based virtual dairy farm devel-

oped by University of Wisconsin is an example of the modern dairy farm management system. The virtual dairy farm collects and integrates the farm's data, streams in real time and then uses AI to analyse that data, which helps farmers make better management decisions.

Intelligent Dairy Farmer's Assistant, or Ida, developed by a Dutch company, uses a motion-sensing device and AI program. The sensor is attached to a cow's neck, which transmits its movements and monitors such real-world behavior as when the cow is chewing cud, lying down, walking, drinking or eating. The system can also detect health issues before they become critical.

Conclusion

Agriculture is one the world's major industries and is indispensable to food security. The world without farmers and agriculture would be a nightmare for the modern world and civilisation. AI holds the promise of driving the agricultural revolution to produce more food using fewer resources. AI, drones and robots are poised to become highly valued tools in the agriculture industry.

AI technology helps farmers react quickly, optimise resources and maximise crop performance, resulting in better yield and food productivity. AI is transforming the agriculture industry in a big way and is one of the key solutions towards improved agricultural productivity. **EFY**

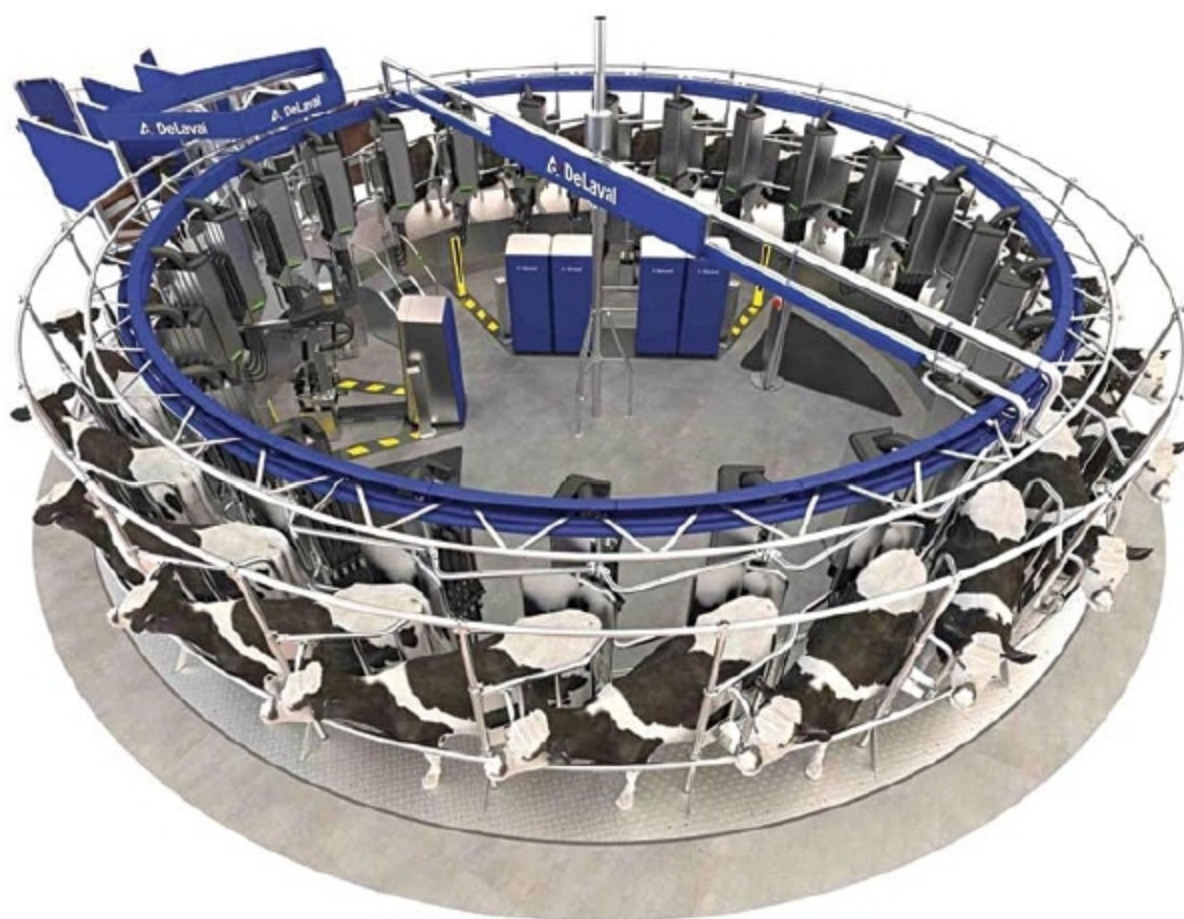


Fig. 4: Prototype of a robotic rotary automatic milking system (Credit: www.foodbev.com)